Submission MS #CHA19-AR-00322

Response to first reviewer's comments

Concerning paper "Generalised early warning signals in multivariate and gridded data with an application to tropical cyclones"

by *J. Prettyman, T. Kuna, V. N. Livina*

We thank the reviewer for the time taken to read through this submission and provide positive constructive feedback with suggestions for improvement. We will reply to the comments here in the order in which they are given.

1. *One of the weaknesses of this work, in my opinion, is a strange motivation of the ACF predictor use in case of tropical cyclones. In Sec. IV this indicator is explained as some approximation of the Jacobian eigenvalue, useful in situation when the real part of this is approaching to zero (bifurcation). But approaching of a cyclone seems much simpler situation when the pressure rapidly decreases without any bifurcation (but due to the specific pressure profile); it can be seen also from the authors' model of the moving cyclone, containing predefined time-dependence without any dynamical system. It is thus unclear how ACF can give more information about the future cyclone than simple pressure observations.*

   reply (1): The motivation of the use to the ACF indicator in the case of tropical cyclones is not based in an understanding that the drop in pressure is caused by a bifurcation in a dynamical system, as may more accurately be the case in the initial formation of a cyclone (as you note), but in the observation that the sudden drop in pressure represents an example of a tipping point in a geophysical system. Indeed, a tipping point is not required to be bifurcational, as noted by [Ashwin et al. Royal Soc Proc A, 2012] who describe noise-induced and rate-induced tipping. Although the tipping impetus is external to the system in this case, we have still classified this as a tipping point.

   The aim of the paper, insofar as we are concerned with the tropical cyclones example, was to apply a tipping point indicator, shown to work well in a bifurcating dynamical system, to a meteorological example of a non-bifurcational tipping point. In this case, the methods do not provide an early warning signal at the 1σ significance level, but in future work we would like to apply these methods to other physical examples of non-bifurcational tipping, if suitable examples can be found.

2. *As I understood, the only situation considered is when an existing cyclone is moving from another place. But practically the more interesting case is when a new cyclone is going to be developed and we need to predict it. Probably, in*

such a case the bifurcation theory could be relevant. Can the authors comment a possibility of such a prediction?

reply (2): We also speculate that the initial formation of a tropical cyclone may be an example of a bifurcation in the atmospheric system, and therefore a good candidate for tipping point analysis. If suitable data can be found, this would be an interesting suggestion for future work in the field.

3. *I think it is needed to briefly describe both the ACR and PS methods in the text (maybe in the supplementary information). This is particularly done for ACR, but only the ACR matrix (Eq. 9) is introduced and the reader has to go to Ref.22 for the method itself.*

   reply (3): We thank you for the suggestion, a brief description of the methods has been included into the manuscript.

4. *There is a problem with EOF indicator applied to two physical variables (pressure and wind speed in this case). When we equally normalise both the time series, we suppose that they have a priori the same significance (prior weights) in the analysis. But, since they are the different physical values, such weights are always unknown and could be selected from some physics only.*

   reply (4): Thank you for the note, we share this concern but, as far as we are aware, a physical basis for the selection of the weights does not exist. We speculate that it is possible to select weights by performing a sensitivity analysis, that is, calculating the tipping point indicator using a range of different relative weightings. In this way, we might expect to see a similar optimal weighting scheme across all examples. A preliminary trial of this did not reveal any clear range of optimal weighting schemes, but we suspect this is because the methods, in the case of the sea-level pressure series, do not provide a significant early warning signal anyway. Possibly better, sub-hourly, data would allow a better comparison.

   We have now included a brief note in the manuscript explaining the possibility of such an analysis. We do not believe there would be significant benefit to fully describing the analysis, but it would be possible to include such in the supplementary material.

5. *Page 2, right column, first par.: "size of the increase in variance" –¿ variance increment?*

   reply (5): We have changed the wording as suggested to make this clearer.

6. *"Indeed, it is not obvious ....". Maybe it is better to say simpler, e.g., the EOF-based leading component may not capture the most interesting component with the largest increase of variance.*

reply (6): We have changed the wording as suggested to make this clearer.

7. *The next par.: "in this case the system spirals towards zero". Not to zero (x can be much greater than 0), but to the process with invariant distribution with finite variance.*

   reply (7): Thank you for catching this. Of course, we are interested in the fact that x has expected value zero, and zero mean in the long term. We have now made this clear in the manuscript.

8. *It is unclear how Eq. 3 is obtained. It follows from Eq. 1 that x_n is completely determined by x_0 and $\eta_0, ...\eta_{n-1}$. But there are both x_0 and infinite history of noises in Eq. 3.*

   reply (8): *We have included two lines of working to make this clear.*

9. *From my view, the estimate Eq. 6 could be obtained in simpler way directly from the Eq. 1 (e.g. by multiplying it by the transpose one and averaging over time).*

   reply (9): We are not sure about how to interpret this advice, in our view there is no simple way to obtain the equation, but we would be interested to discus this.

   - **(10)** *Eq.7: It is not clear how the expansion is done, please include e.g. in supplementary materials.*

   reply (10): Thank you for the suggestion. We have added a little more explanation in the manuscript as to how we arrived at the expression. If it is still thought necessary to provide more detail working through the Taylor expansion of the square root, this could also be included in the supplementary material.

10. *"In this case it is clear that the largest variance...". From my point of view, also a qualitative explanation could be simpler for understanding: if the leading eigenvector (with large enough eigenvalue) of noise covariance matrix (SS^T) is close to orthogonal to the eigenvector of B corresponding to the bifurcating eigenvalue, than the leading principal component PC may not capture the bifurcation.*

    reply (11): Thank you, we have included a note to this effect in the text.

11. *Last par. of Sec. III: It is needed to note that these findings make sense for linear systems only.*

    reply (12): We have now noted the linearity of this particular example at the start of this section, and in the summary.

12. *The sentence after Eq.13 is incorrect: both the stable equilibrium and the limit cycle are possible at zero noise only.*

reply (13): We have now noted that this is strictly true only for zero noise. When there is noise in the system, the equilibrium/limit cycle acts as an attractor to which the system returns after each perturbation.

We hope that the reviewer's concerns have been addressed here and in the revised manuscript. Apparently, the main concern is the first point, that the chosen example of tropical cyclones does not exhibit a bifurcational tipping point. We would like very much to consider another example in future work and hope to be able to investigate the bistability of the Sahara. To this end have been in conversation with Max Plank Institute colleagues to gain rainfall and pollen data from the ECHAM6-HAM2 model, which can be used as a proxy for Saharan greening. Unfortunately, these data are not yet ready for use and not available in public domain. For now, we believe there is merit in applying tipping point analysis to systems exhibiting non-bifurcational tipping, in order to expand this area of research. We also believe there is merit, generally, in the methods presented in this submission, aside from the question of the particular meteorological example used.

We thank the reviewer again for their time in responding carefully to this submission. We hope that the revised manuscript is now suitable for publication in the Chaos magazine.

Yours faithfully,

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